

# FILMREC

A Personalized Recommendation System for Individuals and Groups

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# INTEREST AREA & THE PROBLEM

## MY INTEREST AREA

- Movies: HAD to build something related to movies just because of who I am as a person
- Machine Learning & Recommender Systems: Wanted to explore ML and Rule-Based Recommendation Systems to solve real-world problems
- Social Collaborative Tools: Multiplayer room lobbies like for games (e.g. skribbl)

## THE SPECIFIC PROBLEM

- Choice Paralysis: Excessive time wasted scrolling catalogs instead of watching
- Group Preference Clashes: Diverse, conflicting individual tastes create gridlock in groups
- Rigid Form Constraints: Dropdowns restrict expression; lacks flexible, context-based search

# THE SOLUTION & DEVELOPMENT PATH

## PLANNING BEFORE DEVELOPMENT

Before getting into code, I first designed:

- Database schema
- User flow and application pages
- Recommendation algorithm
- API endpoints and request/response structures
- Overall system architecture

Then I used Antigravity to build it.

## HOW IT WORKS

Users enter their movie preferences.

The application retrieves movie data from the TMDb API.

A custom recommendation engine scores movies based on:

- Genres
- Favourite movies
- Cast & crew
- Runtime
- Free-text preferences

The top recommendations are displayed with compatibility scores and explanations.

## TECH STACK

- **Frontend:** React + TypeScript
- **Backend:** FastAPI (Python)
- **Database:** PostgreSQL (Supabase)
- **Deployment:** Render + Vercel
- **APIs:** TMDb + Groq

# INTERNSHIP LEARNINGS & APPLICATION

## PLANNING BEFORE BUILDING

- I learned the importance of clearly defining the problem statement and deciding the project scope before starting development.
- Instead of coding immediately, I first designed:
  - Database schema
  - Recommendation workflow
  - API structure
  - User interface flow

## USING AI AGENTS EFFECTIVELY

- Defined the project structure
- Refined the design with AI through iterative discussions
- Generated implementation prompts for AntigraVity
- Reviewed and approved every implementation step

## WORKING WITH APIs

I gained hands-on experience with:

- Making API requests
- Parsing JSON responses
- Integrating multiple external APIs
- Connecting the frontend and backend through REST APIs

## DEPLOYING A FULL-STACK APPLICATION

I deployed the complete application using:

- Vercel
- Render
- Supabase

This gave me experience with hosting, environment variables, databases, and production deployment.

# THANK YOU!

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- GitHub Repository: <https://github.com/aditi-kedia/movie-rec-engine>
- Live Demonstration: <https://movie-rec-engine-rouge.vercel.app>
- API Documentation: <https://movie-rec-engine-backend.onrender.com/docs>